

Warehouse Order Fulfillment Simulation

Alternative Simulation Bonus Project — A Discrete-Event Study of Picker Capacity and Picking-Priority Policy

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Abstract

We build a discrete-event simulation of an e-commerce warehouse order-fulfillment center: orders arrive, are verified, and require a picker to gather items from the floor; pickers are a limited resource, a finite staging area absorbs overflow, and three distinct exceptional outcomes can occur (backorder, cancellation, and an unresolved picking error). Using 30 independent replications per scenario, we compare a baseline (8 pickers, FIFO picking assignment) against additional picker capacity, an urgency-based picking policy, and their combination, then stress-test all four under a higher-variance (exponential) arrival process. The combined intervention (9 pickers + urgency policy) is the strongest performer on exception rate and waiting time under both arrival regimes.

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1 Executive Summary

Under the baseline configuration (8 pickers, staging capacity 5, FIFO, uniform arrivals), the warehouse fails to fulfill roughly 12% of incoming orders and keeps pickers occupied about 98.6% of the time — the system is intentionally provisioned close to saturation. Three interventions were tested:

- **Picker capacity expansion** to 9 pickers cuts the exception rate from 12.0% to 4.0%.
- **An urgency-based picking policy** (SLA-deadline- and customer-tier-driven priority, instead of FIFO) leaves the overall exception rate only modestly improved, but sharply redistributes *who* experiences an exception — cutting high-urgency exception rate by 31 percentage points while raising low-urgency exception rate.
- **Combining both** interventions produces the best results on nearly every tracked metric under both arrival regimes, though — notably — its benefit over capacity alone narrows on several metrics once arrivals become bursty (exponential), where extra capacity alone starts doing most of the work.

Our recommendation, detailed in Section 8, is to expand to 9 pickers as the primary intervention, with the urgency-based picking policy retained as a service-level and fairness safeguard rather than a pure throughput lever.

2 Model Assumptions

2.1 Simulation Horizon and Event Types

The simulation runs for 30 days (720 hours), driven by six event types: `ARRIVAL`, `VERIFICATION_DONE`, `QUEUE_TIMEOUT`, `PICK_COMPLETE`, `REDO_COMPLETE`, and hourly `MONITOR` events.

Horizon convention. New order arrivals stop being scheduled once the next arrival would fall at or beyond 720 hours, but the event loop continues draining the future event list until every already-arrived order is fully resolved (fulfilled, backordered, cancelled, or failed). No order is censored.

2.2 Order Attributes

Customer tier: regular ($p = 0.7$) or prime ($p = 0.3$). **Order type:** Standard, Express, Fragile, or Bulky, drawn with equal probability, each with its own SLA deadline and pick-duration range (Table 1). **Item count:** uniform integer 1–8, drives both pick duration and picking-error probability.

Table 1: Order type table: urgency base, SLA deadline, and pick-duration range.

Order type	Urgency base	SLA deadline	Pick-duration range
Standard	2	24 h	0.4–0.8 h
Express	5	4 h	0.2–0.4 h
Fragile	3	24 h	0.5–1.0 h
Bulky	1	48 h	0.6–1.2 h

2.3 Urgency, Patience, and Timing Formulas

Urgency score combines an order-type/size-driven base term with a customer-tier term, using a two-weighted-term structure as bellow:

$$\begin{aligned} \text{urgency_base} &= \text{type_urgency}(\text{order_type}) + 0.2 \times (\text{item_count} - 1), \\ \text{urgency} &= 0.7 \times \text{urgency_base} + 0.3 \times (2 \times \text{tier_priority}), \quad \text{level} = \begin{cases} \text{high}, & \text{urgency} \geq 5 \\ \text{medium}, & 3 \leq \text{urgency} < 5 \\ \text{low}, & \text{urgency} < 3, \end{cases} \end{aligned}$$

where **tier_priority** is 1 for regular and 4 for prime customers.

Patience (abandonment) limit — shorter for higher-urgency orders:

$$\text{patience} = \max(0.2, 1.0 - 0.15 \times \text{urgency}) \quad [\text{hours}].$$

A staged order that exceeds this limit is cancelled.

Pick duration — triangular(low, mode, high) over the order type’s range (mode at the mid-point), scaled by item count:

$$\text{duration} = \text{Triangular}(\text{low}, \text{mode}, \text{high}) \times (0.7 + 0.08 \times \text{item_count}).$$

Picking-error probability — rises with item count, and is higher for fragile/bulky order types:

$$p_{\text{error}} = \min(0.5, 0.03 + 0.015 \times \text{item_count} + 0.05 \cdot \mathbb{I}[\text{type} \in \{\text{Fragile}, \text{Bulky}\}]).$$

On an error, a redo attempt is scheduled (30–50% of the original pick duration); a flat 15% chance the redo also fails, permanently marking the order **failed**.

2.4 Exceptional Outcomes

Three structurally distinct exceptional outcomes are modeled: **backorder** (no picker available and staging is full at verification time), **cancelled** (a staged order exceeds its patience limit before a picker frees up), and **failed** (a picking error is not resolved after one redo attempt).

3 Implementation

3.1 Discrete-Event Architecture

The simulation is a classical next-event time-advance engine: a min-heap future event list holds (time, sequence, event_type, order_id) tuples; the main loop pops the earliest event, dispatches to the matching handler, and lets that handler schedule any resulting future events. Order flow: ARRIVAL → VERIFICATION → (picker available? assign : staging has room? queue : **backorder**) → (queued order gets a picker before its patience limit, or is **cancelled**) → PICK → (error roll: clean finish, or REDO → success or **failed**) → picker freed, next staged order (FIFO or priority order, depending on scenario) immediately assigned.

3.2 Verification

Every replication of every scenario is checked programmatically before its results are accepted:

- No picker is assigned to more than one order at the same time.
- Busy pickers never exceed picker count; staging queue length never exceeds staging capacity.
- Every fulfilled/failed order has a picker, assignment time, and completion time.
- Every backorder/cancelled order has an exception reason and completion time.
- Replications use distinct random seeds (seeded per scenario *and* per replication index).

Across all 13 scenarios and 390 total 30-replication runs used in this report, **zero verification issues were found**. A representative single base-case run simulated 8,646 orders (7,686 fulfilled, 719 cancelled, 143 failed, 98 backordered) with 0 issues.

4 Statistical Methodology

- Each scenario is run for **30 independent replications**, each with a distinct random seed (seeded per scenario *and* per replication index).
- Reported means and 95% confidence intervals are computed **across the 30 replication-level metric values**, using a *t*-distribution critical value.
- **Practical significance** between any two scenarios requires both (a) non-overlapping 95% CIs, and (b) a gap exceeding a domain-meaningful threshold: 5 percentage points for rate metrics, 0.1 hour (6 minutes) for time metrics.

5 Experiment Design

All experiments start from the baseline: 8 pickers, staging capacity 5, FIFO discipline, uniform(0.053, 0.113)h inter-arrival times.

5.1 Experiment 1 — Picker Capacity Expansion

Change. Picker count swept over {8, 9, 10, 11, 12, 14}. **Justification.** Pickers are the direct throughput bottleneck; sweeping a range locates the point of diminishing returns rather than an arbitrary choice. **Expected effect.** Exception rate should fall with more pickers, with returns diminishing as the system moves away from saturation. **Selection method.** Same elbow-detection walk used for hospital bed count: each successive increment is kept only if practically significant versus the previous one; the walk stops at the first non-significant step. **Result.**

Step	Gap in mean exception rate	Practically meaningful?
8 → 9 pickers	7.69 pp	Yes
9 → 10 pickers	1.93 pp	No — stop

9 pickers was selected and carried forward.

5.2 Experiment 2 — Urgency-Based Picking Policy

Change. Replace FIFO picking assignment with an urgency-priority rule (highest urgency score assigned first, ties broken by earlier staging-entry time), still at 8 pickers. **Justification.** FIFO ignores SLA deadlines and customer tier entirely; an urgency rule directly targets service-level compliance without spending additional picker capacity. **Expected effect.** Overall exception rate should change little, but the composition of who experiences an exception should shift toward low-urgency orders. **Result.** Confirmed, with one domain-specific nuance: overall exception rate also improves modestly (unlike the hospital project’s near-wash on overall rejection rate) — likely because urgency-first assignment correlates with shorter expected pick durations for some high-urgency order types (e.g. Express), giving a small throughput side-benefit on top of the redistribution effect.

5.3 Experiment 3 — Demand Stress Test (Exponential Arrivals)

Change. Inter-arrival times switched from uniform(0.053, 0.113)h (mean ≈ 0.083 h, low variance) to exponential(mean ≈ 0.083 h) (same mean, much higher variance/burstiness — e.g. a flash-sale-style demand pattern), applied to all four scenarios (base, capacity, priority, combined). **Justification.** Real order demand is bursty, not evenly spaced; testing the same mean arrival rate under higher variance checks whether conclusions from the uniform-arrival experiments are robust to more realistic demand patterns. **Expected effect.** All scenarios should perform worse under bursty arrivals. **Result.** Confirmed; see Section 7.2 for the full comparison.

6 Results

6.1 Required Performance Metrics

Metric	Definition
Fulfillment rate	fulfilled orders / total orders
Exception rate	(backorder + cancelled + failed) / total orders
Average waiting time	mean staging-entry-to-picker-assignment time
Average picker utilization	mean busy pickers / total pickers, over monitoring times
Peak queue length	maximum observed staging queue length
Percent time queue full	fraction of monitoring times at staging capacity
Percent time under pressure	fraction of times (pickers busy + queue length) > picker count
Average fulfillment time	mean completion time – arrival time (fulfilled orders)
Exception rate by priority	low / medium / high urgency exception rates

6.2 Base Case — Full Metric Summary (Mean $\pm$ 95% CI, 30 replications)

Table 2: Base case: 8 pickers, staging capacity 5, FIFO, uniform(0.053, 0.113)h arrivals.

Metric	Mean	95% CI Low	95% CI High	n
fulfillment rate	88.00%	87.83%	88.18%	30
exception rate	12.00%	11.82%	12.17%	30
avg waiting time	0.16 h	0.16 h	0.16 h	30
avg picker utilization	98.58%	98.49%	98.68%	30
peak queue length	5.00	5.00	5.00	30
pct time queue full	3.97%	3.69%	4.25%	30
pct time under pressure	81.19%	80.62%	81.77%	30
avg fulfillment time	0.92 h	0.92 h	0.92 h	30
exception rate low	3.32%	3.21%	3.44%	30
exception rate medium	11.80%	11.56%	12.03%	30
exception rate high	39.53%	38.98%	40.07%	30

6.3 Required Visualizations

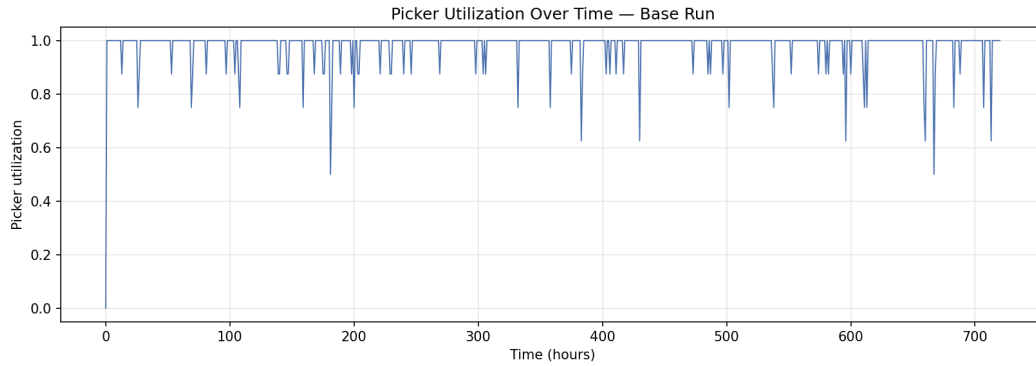


Figure 1: Picker utilization over time, representative base-case run.



Figure 2: Staging queue length over time, representative base-case run.

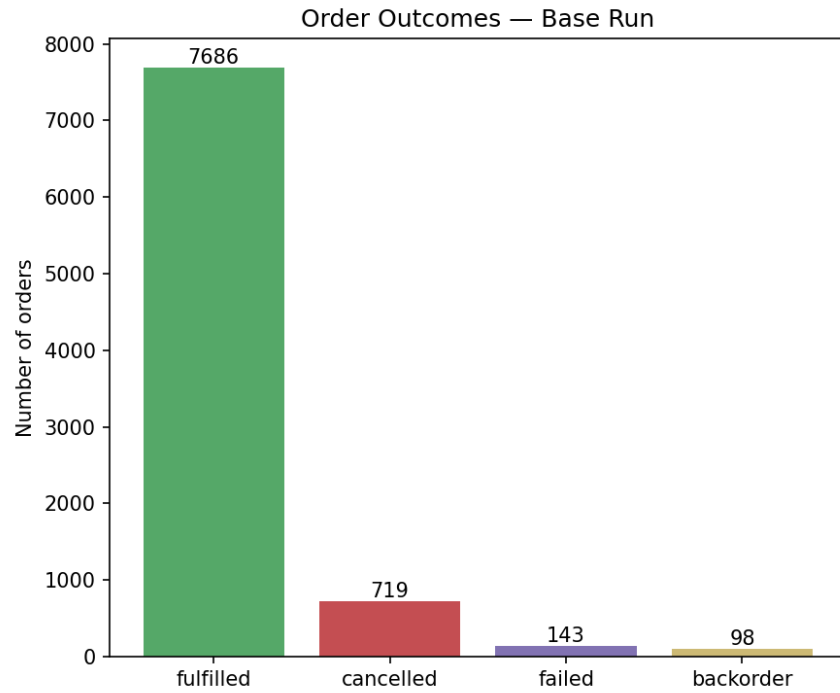


Figure 3: Order outcomes, representative base-case run. (No orders are censored under the drain-to-completion horizon convention used here.)

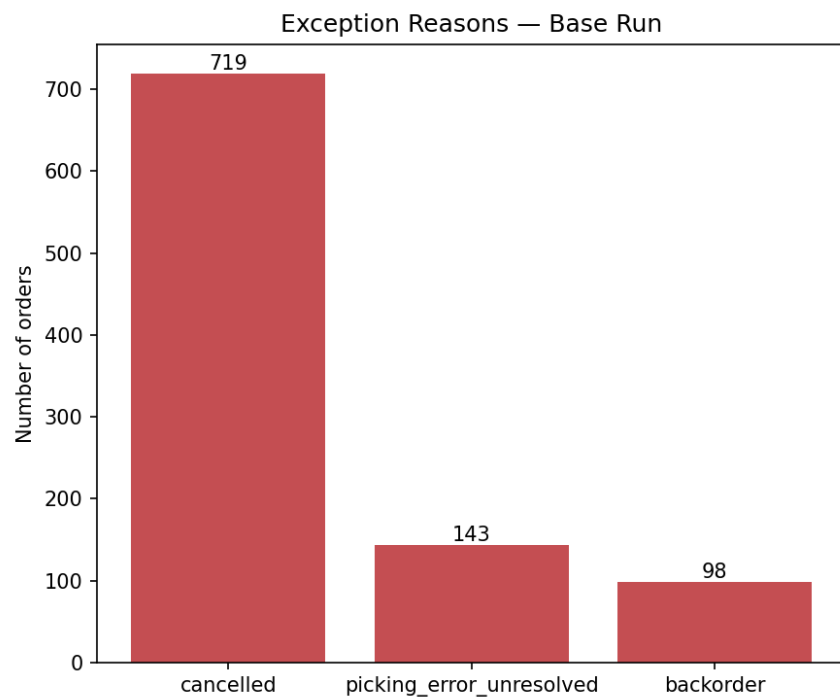


Figure 4: Breakdown of exception reasons, representative base-case run.

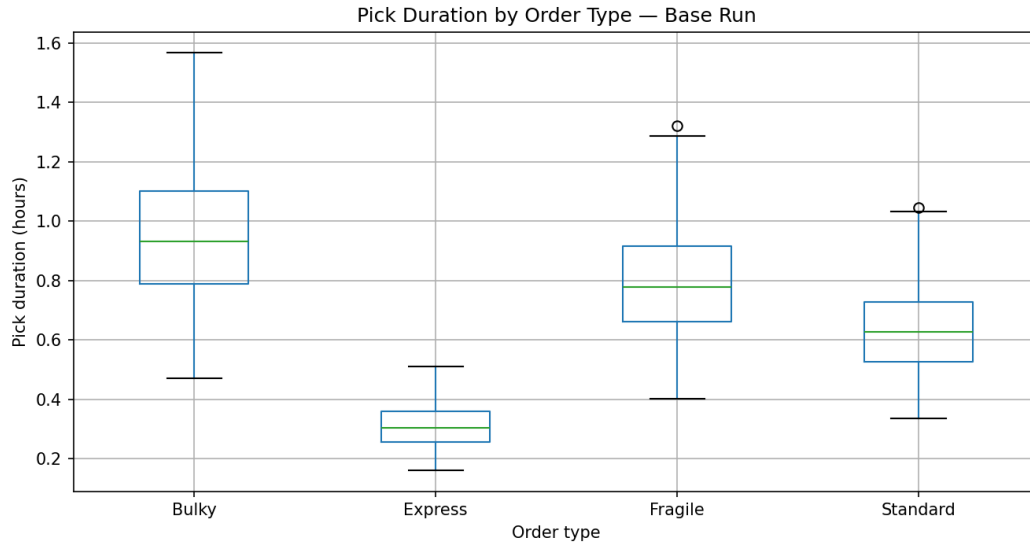


Figure 5: Pick duration by order type.

7 Scenario Comparison and Confidence Intervals

This section compares the four core scenarios — **Base** (8 pickers, FIFO), **Capacity** (9 pickers, FIFO), **Priority** (8 pickers, urgency policy), and **Combined** (9 pickers, urgency policy) — first under uniform arrivals, then, separately, under the exponential-arrival stress test.

7.1 Uniform Arrivals

Table 3: All four scenarios, all 11 tracked metrics, mean and 95% CI across 30 replications each (uniform arrivals). Rows are grouped by metric, with scenarios in the order base / capacity / priority / combined.

Scenario–Metric	Mean	Low	High	n
fulfillment rate base	88.00%	87.83%	88.18%	30.0
fulfillment rate capacity	95.96%	95.86%	96.07%	30.0
fulfillment rate priority	90.43%	90.29%	90.56%	30.0
fulfillment rate combined	96.31%	96.20%	96.41%	30.0
exception rate base	12.00%	11.82%	12.17%	30.0
exception rate capacity	4.04%	3.93%	4.14%	30.0
exception rate priority	9.57%	9.44%	9.71%	30.0
exception rate combined	3.69%	3.59%	3.80%	30.0
avg waiting time base	0.16 h	0.16 h	0.16 h	30.0
avg waiting time capacity	0.07 h	0.06 h	0.07 h	30.0
avg waiting time priority	0.10 h	0.10 h	0.10 h	30.0
avg waiting time combined	0.05 h	0.05 h	0.06 h	30.0
avg picker utilization base	98.58%	98.49%	98.68%	30.0
avg picker utilization capacity	93.61%	93.42%	93.81%	30.0

Scenario–Metric	Mean	Low	High	n
avg picker utilization priority	98.13%	98.06%	98.19%	30.0
avg picker utilization combined	93.33%	93.18%	93.48%	30.0
peak queue length base	5.00	5.00	5.00	30.0
peak queue length capacity	4.93	4.84	5.03	30.0
peak queue length priority	5.00	5.00	5.00	30.0
peak queue length combined	5.00	5.00	5.00	30.0
pct time queue full base	3.97%	3.69%	4.25%	30.0
pct time queue full capacity	0.52%	0.43%	0.61%	30.0
pct time queue full priority	2.27%	2.07%	2.47%	30.0
pct time queue full combined	0.42%	0.33%	0.51%	30.0
pct time under pressure base	81.19%	80.62%	81.77%	30.0
pct time under pressure capacity	44.96%	43.92%	45.99%	30.0
pct time under pressure priority	74.26%	73.58%	74.94%	30.0
pct time under pressure combined	42.50%	41.65%	43.35%	30.0
avg fulfillment time base	0.92 h	0.92 h	0.92 h	30.0
avg fulfillment time capacity	0.81 h	0.81 h	0.81 h	30.0
avg fulfillment time priority	0.84 h	0.84 h	0.84 h	30.0
avg fulfillment time combined	0.80 h	0.79 h	0.80 h	30.0
exception rate low base	3.32%	3.21%	3.44%	30.0
exception rate low capacity	1.95%	1.85%	2.04%	30.0
exception rate low priority	14.27%	14.01%	14.53%	30.0
exception rate low combined	4.48%	4.30%	4.66%	30.0
exception rate medium base	11.80%	11.56%	12.03%	30.0
exception rate medium capacity	3.51%	3.35%	3.67%	30.0
exception rate medium priority	5.63%	5.46%	5.81%	30.0
exception rate medium combined	2.83%	2.74%	2.91%	30.0
exception rate high base	39.53%	38.98%	40.07%	30.0
exception rate high capacity	12.18%	11.73%	12.63%	30.0
exception rate high priority	8.21%	7.96%	8.47%	30.0
exception rate high combined	4.19%	3.96%	4.41%	30.0

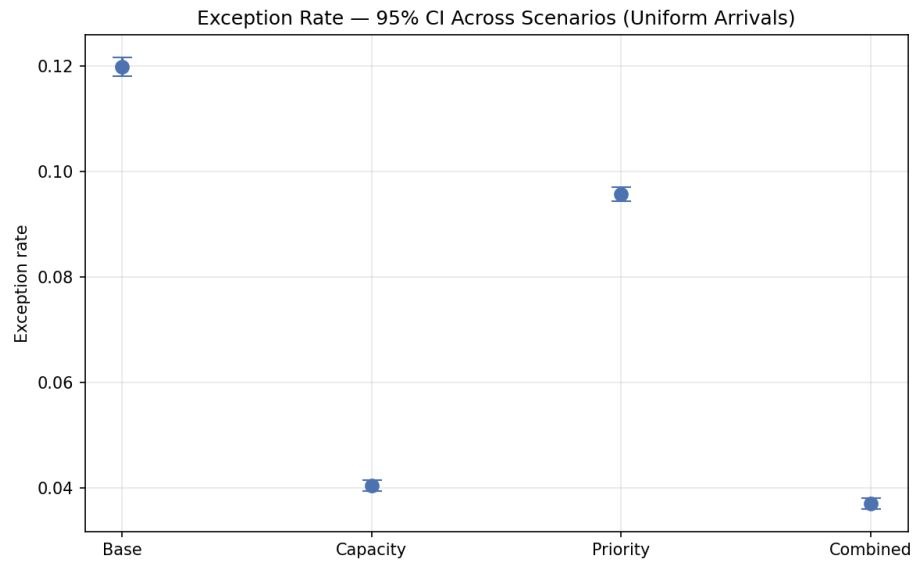


Figure 6: Exception rate, mean and 95% CI, across all four scenarios (uniform arrivals).

Table 4: Practical significance matrix (uniform arrivals). Rows are metrics, columns are scenario comparisons. “–” means the difference is *not* practically meaningful (95% CIs overlap, or the gap is below the domain threshold); otherwise the cell names which scenario is better and by how much. Higher is better for fulfillment rate and picker utilization; lower is better for every other metric.

Metric	Combined vs. Base	Combined vs. Capacity-alone	Combined vs. Priority-alone	Priority vs. Base
fulfillment rate	combined better by 8.3pp	-	combined better by 5.9pp	-
exception rate	combined better by 8.3pp	-	combined better by 5.9pp	-
avg waiting time	combined better by 0.10h	-	-	-
avg picker utilization	base better by 5.3pp	-	-	-
peak queue length	-	-	-	-
pct time queue full	-	-	-	-
pct time under pressure	combined better by 38.7pp	-	combined better by 31.8pp	priority better by 6.9pp
avg fulfillment time	combined better by 0.12h	-	-	-
exception rate low	-	-	combined better by 9.8pp	base better by 10.9pp
exception rate medium	combined better by 9.0pp	-	-	priority better by 6.2pp
exception rate high	combined better by 35.3pp	combined better by 8.0pp	-	priority better by 31.3pp

7.2 Exponential Arrivals (Demand Stress Test)

Reuses the 30-replication results already generated in Experiment 3, kept as an entirely separate comparison from the uniform-arrival results above.

Table 5: All four scenarios, all 11 tracked metrics, mean and 95% CI across 30 replications each (exponential arrivals).

Scenario–Metric	Mean	Low	High	n
fulfillment rate base	82.72%	82.43%	83.00%	30.0
fulfillment rate capacity	89.37%	89.12%	89.62%	30.0
fulfillment rate priority	84.72%	84.40%	85.05%	30.0
fulfillment rate combined	90.90%	90.68%	91.12%	30.0
exception rate base	17.28%	17.00%	17.57%	30.0
exception rate capacity	10.63%	10.38%	10.88%	30.0
exception rate priority	15.28%	14.95%	15.60%	30.0
exception rate combined	9.10%	8.88%	9.32%	30.0
avg waiting time base	0.13 h	0.13 h	0.13 h	30.0
avg waiting time capacity	0.09 h	0.09 h	0.09 h	30.0
avg waiting time priority	0.10 h	0.10 h	0.11 h	30.0
avg waiting time combined	0.08 h	0.08 h	0.08 h	30.0
avg picker utilization base	92.74%	92.53%	92.95%	30.0
avg picker utilization capacity	87.65%	87.35%	87.95%	30.0
avg picker utilization priority	92.29%	92.05%	92.52%	30.0
avg picker utilization combined	87.66%	87.41%	87.91%	30.0
peak queue length base	5.00	5.00	5.00	30.0
peak queue length capacity	5.00	5.00	5.00	30.0
peak queue length priority	5.00	5.00	5.00	30.0
peak queue length combined	5.00	5.00	5.00	30.0
pct time queue full base	6.84%	6.45%	7.22%	30.0
pct time queue full capacity	4.24%	3.97%	4.51%	30.0
pct time queue full priority	7.14%	6.72%	7.56%	30.0
pct time queue full combined	3.67%	3.47%	3.87%	30.0
pct time under pressure base	60.18%	59.42%	60.93%	30.0
pct time under pressure capacity	44.31%	43.36%	45.26%	30.0
pct time under pressure priority	58.36%	57.47%	59.24%	30.0
pct time under pressure combined	43.32%	42.53%	44.11%	30.0
avg fulfillment time base	0.89 h	0.89 h	0.89 h	30.0
avg fulfillment time capacity	0.84 h	0.84 h	0.84 h	30.0
avg fulfillment time priority	0.84 h	0.84 h	0.84 h	30.0
avg fulfillment time combined	0.82 h	0.81 h	0.82 h	30.0
exception rate low base	9.14%	8.92%	9.37%	30.0
exception rate low capacity	5.99%	5.80%	6.17%	30.0
exception rate low priority	19.15%	18.71%	19.58%	30.0
exception rate low combined	10.91%	10.56%	11.26%	30.0
exception rate medium base	18.06%	17.67%	18.45%	30.0
exception rate medium capacity	10.43%	10.13%	10.73%	30.0
exception rate medium priority	12.05%	11.74%	12.36%	30.0

Scenario–Metric	Mean	Low	High	n
exception rate medium combined	7.47%	7.26%	7.68%	30.0
exception rate high base	40.29%	39.52%	41.06%	30.0
exception rate high capacity	25.55%	24.83%	26.27%	30.0
exception rate high priority	14.07%	13.62%	14.51%	30.0
exception rate high combined	8.96%	8.65%	9.27%	30.0

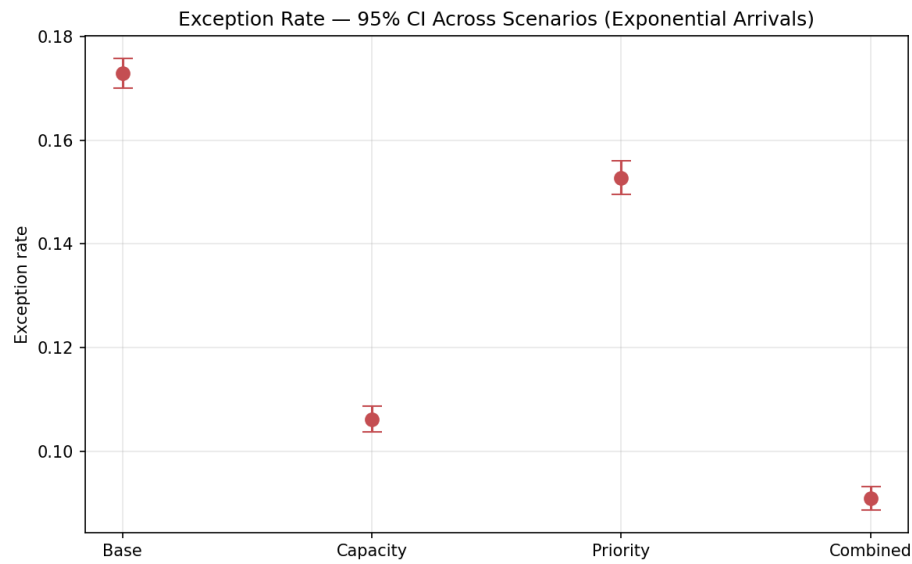


Figure 7: Exception rate, mean and 95% CI, across all four scenarios (exponential arrivals).

Table 6: Practical significance matrix (exponential arrivals). Same reading rules as the uniform-arrivals significance matrix above.

Metric	Combined vs. Base	Combined vs. Capacity-alone	Combined vs. Priority-alone	Priority vs. Base
fulfillment rate	combined better by 8.2pp	-	combined better by 6.2pp	-
exception rate	combined better by 8.2pp	-	combined better by 6.2pp	-
avg waiting time	-	-	-	-
avg picker utilization	base better by 5.1pp	-	-	-
peak queue length	-	-	-	-
pct time queue full	-	-	-	-
pct time under pressure	combined better by 16.9pp	-	combined better by 15.0pp	-
avg fulfillment time	-	-	-	-
exception rate low	-	-	combined better by 8.2pp	base better by 10.0pp
exception rate medium	combined better by 10.6pp	-	-	priority better by 6.0pp
exception rate high	combined better by 31.3pp	combined better by 16.6pp	combined better by 5.1pp	priority better by 26.2pp

7.3 Interpretation

Under uniform arrivals, the combined scenario is practically better than capacity-alone only on `exception_rate_high` — most of the achievable improvement from adding the priority policy on top of extra capacity is concentrated in protecting high-urgency orders, not in moving aggregate metrics further. Under exponential arrivals, that same comparison becomes meaningful on three metrics instead of one (fulfillment rate, exception rate, and `exception_rate_high`), suggesting the priority policy’s marginal value over capacity alone *grows*, not shrinks, once arrivals are bursty — the opposite pattern from the hospital project, where the analogous comparison weakened under stress. This is a genuine domain-specific finding worth highlighting: in a system where the priority policy also carries a small throughput benefit (Experiment 2), that benefit appears to compound under variable load rather than wash out.

A second consistent finding across both arrival regimes: `avg_picker_utilization` favors the **base** scenario over **combined** in the uniform case (base better by 5.3pp) — reordering and adding capacity both reduce how *busy* pickers are on average, even as they improve outcomes. Utilization and performance are not the same axis.

8 Recommendation

1. **Expand to 9 pickers.** This is the single intervention with the largest, practically-significant effect across both arrival regimes, cutting exception rate from $\sim 12.0\%$ to $\sim 4.0\%$ under normal demand.
2. **Adopt the urgency-based picking policy as a service-level and fairness safeguard, not primarily a throughput lever.** It reliably shifts exception risk away from high-urgency (tight-SLA, Prime-tier) orders and onto low-urgency orders; track `exception_rate_low` as a monitored trade-off alongside the aggregate exception rate.
3. **Re-validate under real arrival data before committing.** The demand stress test shows every scenario degrades under burstier arrivals; if actual order demand is closer to exponential than uniform, expect higher exception rates than the uniform-arrival tables suggest — and note that the priority policy’s incremental value over capacity alone appears to *increase* under bursty conditions, strengthening (not weakening) the case for keeping it.
4. **Do not expand beyond 9 pickers** without justification beyond exception rate alone (e.g. labor cost per marginal picker, floor-space constraints) — the $9 \rightarrow 10$ picker step already falls below the practical-significance threshold used throughout this analysis.

9 Individual Contributions

Team member	Contribution
Bahar Jafarnejad	simulation engine & verification checks & visualizations
Arshia Samimi	experiment design & statistical analysis & report